Low-energy matched caps and gauge-free first-order stability

28 September 2026. Continuation of I44 and I52, after the marked-center construction. All statements below concern actual smooth metrics.

The local construction fills a nearly Euclidean annulus at a cost controlled by its full curvature energy and its zeroth-order deviation. It preserves the original outer collar exactly. Applied on sufficiently long annular ranges, it removes the gauge and scalar-curvature hypotheses from the lower first-order bound near selected recovery metrics. It does not parametrize arbitrary near-minimizers.

1. A matched cap from zeroth-order closeness and critical energy

Let e denote the Euclidean metric and $\mathscr{A} = \{1 < |x| < 4\} \subset \mathbb{R}^4$. The curvature norm can use either standard tensor convention; the universal constants absorb that fixed choice.

Theorem 1 (small-energy matched cap). There are $a_0, \varepsilon_0, C > 0$ such that every smooth metric g on $\mathscr{A}$ with

$$a = \|g - e\|_{L^\infty(e)} \leqslant a_0, \qquad E_0 = \int_{\mathscr{A}} |\operatorname{Rm}_g|^2 \, dV_g \leqslant \varepsilon_0$$

has a smooth extension G to B_4, equal to g on $\{7/2 < |x| < 4\}$, satisfying

$$\int_{B_4} |\operatorname{Rm}_G|^2 \, dV_G \leqslant C(E_0 + a^2). \tag{1.1}$$

There is no derivative bound on the input. No boundary diffeomorphism is needed. In particular (1.1) bounds the self-dual Weyl cost.

The proof uses a localized Ricci–DeTurck deformation. The metric is not required to solve any stationary equation. The small C^0 hypothesis supplies the parabolic and Sobolev control; we do not invoke curvature-only harmonic-coordinate regularity.

1.1 Regularization and a uniform smoothing time

Embed B_5 in a fixed flat torus. Extend g as $g_0 = e + \zeta(g - e)$, where ζ is supported in $\mathscr{A}$ and equals one on

$$\Omega = \{9/8 < |x| < 15/4\}.$$

This is a smooth positive extension with the same small supremum bound. Its curvature away from Ω can be arbitrarily large; no estimate below charges that curvature.

Choose a fixed smooth radial cutoff with

$$\operatorname{supp}\chi \subset \{5/4 < |x| < 13/4\}, \qquad \chi = 1 \text{ on } \{7/4 < |x| < 9/4\}, \qquad 0 \leqslant \chi \leqslant 1.$$

For $0 < \eta < 1$, put $q = \chi^2 + \eta^2$, and solve

$$\partial_t g_\eta = -2q \operatorname{Ric}(g_\eta) + \mathscr{L}_{qW(g_\eta)} g_\eta, \qquad W^k(g) = g^{ab}\Gamma(g)^k_{ab}, \qquad g_\eta(0) = g_0. \tag{1.2}$$

The background connection is the flat one. The full Lie derivative $\mathscr{L}_{qW}$, not $q\mathscr{L}_W$, is important both for the coordinate equation and the geometric energy identity. For every fixed $\eta > 0$, this is a smooth strictly parabolic system. Local existence is the classical DeTurck reduction and quasilinear parabolic existence theorem.

Here is the uniform continuation estimate needed for the proof. Write $u = g_\eta - e$. In flat coordinates (1.2) takes the form

$$\partial_t u = q\, g_\eta^{ab}\partial_{ab}u + q\,\mathscr{E}(g_\eta, Du, Du) + Dq\cdot\mathscr{B}(g_\eta)Du, \tag{1.3}$$

with smooth bounded algebraic coefficients while $|u|$ is small. Set

$$L = \partial_t - q g_\eta^{ab}\partial_{ab}, \qquad v = |u|_e^2, \qquad w = q|Du|_e^2.$$

The fixed cutoff gives

$$|Dq|^2/q + |D^2 q| \leqslant C_\chi$$

uniformly in η. Differentiating (1.3), using ellipticity near e, and absorbing mixed Hessian terms by Young's inequality gives

$$Lv \leqslant -cw + Cv, \qquad Lw \leqslant -cq^2|D^2u|^2 + Cw^2 + Cw. \tag{1.4}$$

For clarity, differentiation of the principal coefficient contributes $qDq\,Du\,D^2u$ and $q^2(Du)^2D^2u$. Their costs are bounded by a fraction of $q^2|D^2u|^2$ plus $Cw + Cw^2$. The differentiated cutoff drift contributes $D^2q\,w$ and $(|Dq|/\sqrt{q})w^{3/2}$, bounded by the same right side. No third derivative of q or input derivative bound is used. In the first inequality the nonlinear term is $O(|u|w)$; choosing a_0 small absorbs it.

Fix $T > 0$ independent of the input, and take $K > 2e^{CT}$. Then $v \leqslant a^2e^{Ct}$ while the solution exists. At a spatial maximum of $F = w/(Ka^2 - v)$, the quotient gradient terms cancel, and (1.4) implies

$$LF \leqslant \{C(Ka^2 - v) - c\}F^2 + CF \leqslant -c'F^2 + CF. \tag{1.5}$$

After reducing a_0, scalar comparison yields

$$\|u(t)\|_\infty \leqslant Ca, \qquad q|Du(t)|^2 \leqslant Ca^2/t \quad (0 < t \leqslant T). \tag{1.6}$$

The case $a = 0$ is the stationary flat solution.

For fixed η, the second estimate is an ordinary gradient bound on every $[\delta, T]$, because $q \geqslant \eta^2$. Together with uniform ellipticity and small $\|u\|_\infty$, smooth parabolic continuation extends this solution to the same time T. The constants for higher global derivatives may depend on η; that is harmless. On compact subbands where $\chi = 1$, interior parabolic estimates applied to (1.3) instead give

$$\|g_\eta(T) - e\|_{C^2(\{15/8<|x|<17/8\})} \leqslant Ca, \tag{1.7}$$

uniformly in η. One first uses (1.6) on $[T/2, T]$; rescaling u by a gives bounded coefficients and then the usual interior bootstrap. This explains the linear dependence on a in (1.7), not just a bound independent of the input.

1.2 Keeping the actual outer collar

For a fixed smooth g_0, let $H(s)$ be its ordinary smooth Ricci–DeTurck flow on its initial existence interval. For $\eta^2 T$ in that interval, $h_\eta(t) = H(\eta^2 t)$ tends to g_0 in C^∞, uniformly for $0 \leqslant t \leqslant T$. On any collar U where $\chi = 0$, both g_η and h_η solve the same η^2-speed DeTurck equation.

Their difference z, initially zero, satisfies

$$\partial_t z = \eta^2 g_\eta^{ab}\partial_{ab}z + \eta^2\mathscr{E}(g_\eta, Dz, Dz) + \eta^2 B_{g_0}Dz + \eta^2 C_{g_0}z. \tag{1.8}$$

The last two coefficients, also depending algebraically on g_η, z, are bounded by constants of this fixed smooth datum, uniformly for small η; this follows by expanding $Dg_\eta = Dh_\eta + Dz$. Smallness of z absorbs its quadratic gradient term in the equation for $|z|^2$. Therefore

$$(\partial_t - \eta^2 g_\eta^{ab}\partial_{ab})|z|^2 \leqslant C_{g_0}\eta^2|z|^2.$$

Exponential barriers from the two boundary components of U give, on every smaller collar at distance $d > 0$,

$$|z| \leqslant C_{g_0}\exp[-cd^2/(\eta^2 t)], \qquad 0 < t \leqslant T. \tag{1.9}$$

For example use barriers $e^{-\lambda r + C\eta^2\lambda^2 t + C_{g_0}\eta^2 t}$ in the radial distance, and optimize λ. Terms from the radial Hessian are absorbed by increasing C.

Interior parabolic estimates on spatial scale $\eta\sqrt{t}$, followed by differentiation on smaller collars, multiply (1.9) only by powers of $(\eta\sqrt{t})^{-1}$. The rescaled coefficient bounds follow from (1.6) and interior bootstrapping; the coefficients involving h_η are smooth for this fixed datum. Thus, for each fixed integer m,

$$g_\eta \longrightarrow g_0 \quad \text{in } C^m \text{ on smaller } \{\chi = 0\} \text{ collars, uniformly on } [0, T]. \tag{1.10}$$

Exponential decay also handles $t \downarrow 0$. No smooth limit through the degeneracy interface is asserted. Only collars a positive distance from $\operatorname{supp}\chi$ are used.

There is an independent classical check of this collar step. On a ball inside such a buffer put $s = \eta^2 t$. The rescaled flow is the ordinary small-C^0 DeTurck flow. Schnürer–Schulze–Simon, Theorem 4.4 gives interior derivatives up to the initial time, depending on the fixed smooth initial metric, not on boundary derivatives. Integrating the local equation gives $O_{g_0}(\eta^2 T)$ convergence in each fixed interior C^m norm. This checks the uniform-in-time version of (1.10); it is not used to assert a global smooth degenerate flow.

1.3 Critical energy is retained in the annulus

Put $E_\eta(t) = \int_\Omega |\operatorname{Rm}(g_\eta)|^2$. The geometric curvature variation for $-2q\operatorname{Ric}$ has the structure

$$q\Delta\operatorname{Rm} + \nabla q * \nabla\operatorname{Rm} + \nabla^2 q * \operatorname{Rm} + q\operatorname{Rm} * \operatorname{Rm}.$$

It is the localized Ricci curvature equation of Yang, Section 9, equation (9.7). Here only that equation is used, not his theorem's additional L^p Ricci hypothesis. The Lie derivative in (1.2) contributes only boundary transport to the integral.

Integrate the $\nabla^2 q$ term by parts. Metric compatibility then gives, with $D_\eta = \int_\Omega q|\nabla\operatorname{Rm}|^2$,

$$E_\eta' \leqslant -cD_\eta + CE_\eta + C\int_\Omega q|\operatorname{Rm}|^3 + B_\eta. \tag{1.11}$$

Every mixed derivative is absorbed using $|\nabla q|^2/q \leqslant C$, which follows from the coordinate cutoff bound and zeroth-order metric equivalence. In particular no bound on the metric Hessian of χ is assumed. Near $\partial\Omega$, $q = \eta^2$ and $Dq = 0$. Equation (1.10), with $m = 3$, bounds all diffusion and transport fluxes there, so

$$|B_\eta| \leqslant C_{g_0}\eta^2. \tag{1.12}$$

Large curvature of the arbitrary extension outside Ω does not enter this inequality.

One may alternatively estimate the Hessian term directly: $|\nabla_g^2 q| \leqslant C(1+a/\sqrt{t})$ by (1.6). This replaces CE_η in (1.11) by $C(1+a/\sqrt{t})E_\eta$. Its uniformly integrable time coefficient gives the same conclusion below, without that integration by parts.

The inhomogeneous Sobolev inequality on the fixed annulus is uniform because g_η stays uniformly close to e. Apply it and Kato's inequality to $\sqrt{q}\,|\operatorname{Rm}|$:

$$\|\sqrt{q}\,|\operatorname{Rm}|\|_4^2 \leqslant C(D_\eta + E_\eta), \qquad \int_\Omega q\,|\operatorname{Rm}|^3 \leqslant C\sqrt{E_\eta}(D_\eta + E_\eta). \tag{1.13}$$

No zero boundary values are needed in this inhomogeneous inequality. Choose ε_0 small and then η small for the fixed datum. A continuity bootstrap absorbs the first term on the right of (1.13) into (1.11). Gronwall gives

$$E_\eta(T) \leqslant CE_0 + C_{g_0}\eta^2. \tag{1.14}$$

Both the time and C are independent of input derivatives.

1.4 Finishing the smooth cap

On $15/8 < |x| < 17/8$, interpolate from e on the inside to $g_\eta(T)$ on the outside using a fixed smooth cutoff. Equation (1.7) bounds its full curvature cost by Ca^2. Use e further inside and $g_\eta(T)$ up to radius $27/8$.

On $27/8 < |x| < 7/2$, interpolate back to the original g, then use g for $7/2 < |x| < 4$. This second collar lies in the region of (1.10). Its metric converges in C^2 to the original one for this fixed datum, so its curvature integral equals the original collar integral plus $o_\eta(1)$. Convex interpolation preserves positivity.

The resulting smooth cap has total cost

$$C(E_0 + a^2) + C_{g_0}\eta^2 + o_\eta(1).$$

If $E_0 + a^2 > 0$, choose η so that the last errors are at most $E_0 + a^2$. This choice may depend on every derivative of the individual input, but the final bound does not. If both quantities are zero, the input is e and the flat cap suffices. This proves Theorem 1.

2. Near-Einstein energy supplies the global curvature budget

For a closed oriented four-manifold set

$$\mathscr{E}(g) = \inf_{\lambda \neq 0} \frac{\int |\operatorname{Ric}_g - \lambda g|^2\, dV_g}{\lambda^2 V_g}, \qquad V_g = \operatorname{Vol}_g(M),$$

and write

$$P = \int |\operatorname{Ric}^\circ|^2, \quad S = \int s^2, \quad R = P + S/4, \quad L_s = S - \frac{(\int s)^2}{V} \geqslant 0.$$

In the actual positive-budget family $R > 0$, and $\mathscr{E} = (4P + L_s)/R$. Hence $P \leqslant \mathscr{E}R/4$. The Chern–Gauss–Bonnet identity gives

$$8\pi^2\chi = \int (|W^+|^2 + |W^-|^2) + R/6 - 2P/3 \geqslant (1 - \mathscr{E})R/6. \tag{2.1}$$

Consequently, if $\mathscr{E} < 1/2$,

$$R \leqslant 96\pi^2\chi, \qquad \int |\operatorname{Rm}|^2 \leqslant C\chi. \tag{2.2}$$

The second estimate follows by inserting this bound into the same identity and the curvature decomposition. These bounds need neither scalar normalization nor a volume estimate.

On the balanced actual family

$$M_j = X_j \# k_j \overline{\mathbb{CP}}^2 \# \ell_j (S^1 \times S^3), \quad A_j = c_1^2(X_j), \quad Q_j = k_j + 4\ell_j, \quad 0 < h_j = Q_j/A_j \to 0,$$

one has $\chi(M_j) = A_j/3 + k_j - 2\ell_j = O(A_j)$. Thus every sequence with bounded $\mathscr{E}/h_j$ automatically has full curvature energy $O(A_j)$.

Here X_j has its complex orientation and is a finite cover of the fixed closed complex-hyperbolic surface used in I44/I52. The blowup and handle class supply is the inherited actual one of I03, not a formal vector on a cap. Its two classical all-metric budgets used below are, explicitly,

$$S \geqslant 32\pi^2 A, \qquad R \geqslant 8\pi^2 A(1 + h).$$

They are the scalar and Ricci-energy inequalities detailed in I44, Sections 5–6, with classical input from LeBrun. No monopole supply is demanded on the auxiliary capped manifold.

3. Long marked annuli force the optimal first-order lower bound

Theorem 2 (annular first-order stability). On the actual family above, suppose that around every marked blowup core there are L_j disjoint fixed-ratio annular bands, all core families mutually disjoint. Each band has marked coordinates, identifying its filled inside with the corresponding blowdown ball. It encloses only its designated blowup, not a handle or another core. Assume, after scaling each band to $\mathscr{A}$, that

$$\max_{i,\nu} \|g_j - e\|_{C^0(\mathscr{A}_{i,\nu,j})} = a_j \longrightarrow 0, \qquad L_j h_j \longrightarrow \infty. \tag{3.1}$$

Then

$$\liminf_j \frac{\mathscr{E}(g_j)}{h_j} \geqslant 4. \tag{3.2}$$

There is no scalar, gauge, derivative, or stationary assumption. Away from these annuli the metric is arbitrary.

Proof. If the ratio has no bounded subsequence there is nothing to prove. On any bounded subsequence, (2.2) supplies total curvature energy $O(A_j)$. Choose a minimum-energy band for each blowup. Their total energy satisfies

$$\sum_i E_{i,j} \leqslant C A_j / L_j. \tag{3.3}$$

Call a selected band bad if $E_{i,j} > \varepsilon_0$, and let b_j be the number of bad blowups. Then $b_j \leqslant C A_j/(L_j \varepsilon_0) = o(Q_j)$. Theorem 1 applies to every good band. Replace only those insides by matched balls; all handles and bad blowups remain unchanged. The resulting closed manifold is $N_j = X_j \# b_j \overline{\mathbb{CP}}^2 \# \ell_j (S^1 \times S^3)$. The sum of self-dual Weyl cap costs is

$$C_{+,j} \leqslant C\left(A_j/L_j + k_j a_j^2\right) = o(Q_j). \tag{3.4}$$

Indeed $A_j/(L_j Q_j) = 1/(L_j h_j) \to 0$ and $k_j \leqslant Q_j$. No maximum individual energy needs to tend to zero.

The partial-blowdown version of I44, Theorem 5 retains the cost of the b_j unremoved blowups. Suppressing j, the signature formula and the original scalar identity give

$$\begin{aligned} B_M &= 12\pi^2(k-b) + B_N + K_+ - C_+, \\ P_M &= 8\pi^2(Q-6b) + (S_M - S_0)/12 + 4(B_N + K_+ - C_+), \end{aligned} \tag{3.5}$$

where $S_0 = 32\pi^2 A$, $B_N = \int_N |W_N^-|^2$, and K_+ is the positive-Weyl energy of the removed cores. Consequently

$$\begin{aligned} &(1+h)R_M\left(\mathscr{E}_M - \frac{4h}{1+h}\right) \\ &\quad = \frac{1-3h}{3}(S_M - S_0) + 16(B_N + K_+ - C_+) + (1+h)L_s - 192\pi^2 b. \end{aligned} \tag{3.6}$$

The actual scalar/projection budget on M gives $S_M \geqslant S_0$ and $R_M \geqslant 8\pi^2 A(1+h)$. For $h \leqslant 1/3$ all terms except the two displayed costs are nonnegative. Hence

$$\frac{\mathscr{E}(g_j)}{h_j} \geqslant \frac{4}{1+h_j} - \frac{24b_j + 2C_{+,j}/\pi^2}{Q_j(1+h_j)^2}. \tag{3.7}$$

The two costs are $o(Q_j)$, proving (3.2). No conformal reduction occurred. The cap is an auxiliary smooth metric, so it need not carry a new monopole solution. The original actual supplier remains on M_j, exactly as required by I44. QED.

The proof gives a finite, all-metric estimate, not only a sequential statement. For $0 < h \leqslant 1/7$, $a \leqslant a_0$, $L \geqslant 1$, and χ/A bounded as on this family,

$$\boxed{\mathscr{E}(g) \geqslant \frac{4h}{1+h} - C\left(\frac{1}{L} + \frac{k}{A}a^2\right).} \tag{3.8}$$

If $\mathscr{E} < 1/2$, the proof just given applies with $b \leqslant CA/(L\varepsilon_0)$, whether or not A/L is small. If $\mathscr{E} \geqslant 1/2$, (3.8) follows immediately from $4h/(1+h) \leqslant 1/2$. This explains why the competitive-energy premise is not an assumption of the result.

There is a useful robustness form. Leave r marked blowups without any prescribed annular family, and let a_i be the maximal amplitude over the L bands of each remaining blowup. Discard also those with $a_i > a_0$, at most $a_0^{-2}\sum_i a_i^2$ of them. The same partial-cap proof gives

$$\mathscr{E}(g) \geqslant \frac{4h}{1+h} - C\left(\frac{r}{A} + \frac{1}{L} + \frac{1}{A}\sum_i a_i^2\right). \tag{3.8a}$$

Thus $r = o(Q)$, $\sum_i a_i^2 = o(Q)$, and $Lh \to \infty$ suffice for (3.2). A maximum amplitude tending to zero at every core is not necessary. The unstructured regions are retained, not silently filled, and their number is charged in the identity.

One can average amplitudes over the bands as well. If $a_{i\nu}$ is the amplitude of band ν at core i, select the band minimizing $E_{i\nu} + a_{i\nu}^2$. The sum of selected scores is at most $CA/L + L^{-1}\sum_{i,\nu} a_{i\nu}^2$. Discard scores larger than $\min\{\varepsilon_0, a_0^2\}$; the others satisfy both cap hypotheses. Thus (3.8a) improves to

$$\mathscr{E}(g) \geqslant \frac{4h}{1+h} - C\left(\frac{r}{A} + \frac{1}{L} + \frac{1}{AL}\sum_{i,\nu} a_{i\nu}^2\right). \tag{3.8b}$$

This is the same partial-cap argument, not a regularity assertion for the discarded bands.

Corollary 3 (arbitrary C^0-small perturbations of selected recoveries). The CH shrinking-surgery bases can be selected so that every sequence of smooth metrics g_j satisfying

$$\|g_j - g_j^0\|_{L^\infty(g_j^0)} \to 0$$

obeys (3.2). This removes the TT, Bianchi, constant-scalar and amplitude-rate assumptions from the lower-bound part of I52.

Proof. At each fixed j, choose L_j with $L_j h_j \to \infty$. Inside each disjoint CH blowup chart choose a finite annular range of that many bands, with outer radius so small that the scaled CH deviation on every band is at most a prescribed $\varepsilon_j \to 0$. Keep its innermost radius positive at this fixed j, and then take the actual donor scale sufficiently smaller. The exterior Yamabe convergence of I52 is uniform on this fixed finite union, so the normalized recovery is Euclidean-close there. Choose the donor scales to make that additional error at most ε_j, along with all previous recovery selections. Now take the outer j-diagonal.

This order of choices matters. No control of the Yamabe factor on every rescaled donor is inferred from exterior convergence. For the stated competitors, the annular amplitude in (3.1) is at most $C(\varepsilon_j + \|g_j - g_j^0\|_\infty)$, which tends to zero. Theorem 2 applies. QED.

The statement is invariant under simultaneous pullback. Thus a competitor which can be made C^0-close to these bases by any orientation-preserving diffeomorphism is also covered; no bound on that diffeomorphism is needed. The compact moving-center selection from the companion proof remains available by choosing the finite annular ranges uniformly at each fixed compact configuration set. The old exact-slice quadratic remainder and equality characterization remain separate stronger conclusions inside their specified slices.

3.1 The minimum over every shrinking zeroth-order neighborhood

This extension determines an actual variational infimum, not only the energy of a prescribed perturbation. Let $b_j \to 0$ be the maximum Euclidean error of the selected base on its marked annuli. For small fixed $\rho_0 > 0$, define

$$m_j(\rho) = \inf\left\{ \frac{\mathscr{E}(g)}{h_j} : g \text{ smooth}, \ \|g - g_j^0\|_{L^\infty(g_j^0)} \leqslant \rho \right\}, \qquad 0 \leqslant \rho \leqslant \rho_0.$$

The amplitude on each band is at most $b_j + (1 + b_j)\rho$. Equation (3.8) and $k_j \leqslant Q_j$ give

$$\frac{4}{1 + h_j} - C\left((L_j h_j)^{-1} + (b_j + \rho)^2\right) \leqslant m_j(\rho) \leqslant \frac{\mathscr{E}(g_j^0)}{h_j}. \tag{3.9}$$

The right side tends to four by the actual recovery construction. It follows for every radius sequence $\rho_j \to 0$, with no rate relative to h_j, that

$$\lim_j m_j(\rho_j) = 4. \tag{3.10}$$

Also both iterated lower/upper limits as $\rho \downarrow 0$ equal four. No existence of the inner j-limit for a fixed positive ρ is asserted: (3.9) gives only the interval $[4 - C\rho^2, 4]$ in that case.

For the intrinsic distance

$$d_j(g) = \inf_{\varphi \in \mathrm{Diff}^+(M_j)} \|\varphi^* g - g_j^0\|_{L^\infty(g_j^0)},$$

approximate the infimum by diffeomorphisms and use invariance of $\mathscr{E}$. The same lower bound holds with $\rho = d_j(g)$ whenever this distance is small. In particular, for every fixed $\delta > 0$, a subsequence satisfying $\mathscr{E}(g_j)/h_j \leqslant 4 - \delta$ must eventually satisfy

$$d_j(g_j) \geqslant c \min\{\rho_0, \sqrt{\delta}\} > 0. \tag{3.11}$$

The vanishing $b_j, (L_j h_j)^{-1}, h_j$ errors are absorbed first. This is separation from the selected recovery orbits, not a classification of such a putative new geometry. It also does not prove that every coefficient-four sequence approaches those orbits. The quadratic error in (3.9) is a proved bound, not a claim of optimal amplitude dependence.

4. What changed and what remains

The matched cap is stronger than the previously requested vanishing modulus: its cost is quadratic in the zeroth-order amplitude plus the actual curvature energy. The localized flow does not need to preserve the gauge fields or converge to an Einstein metric. It constructs a temporary filling metric used in an exact topological identity. This distinction avoids imposing unnecessary flow conditions.

The fixed-base exact-gauge obstruction in X09 therefore does not block the first-order estimate. The intrinsic cap argument covers arbitrary C^0-nearby competitors without solving that gauge. The unrestricted infimum over every metric on M_j is still not computed: no theorem here supplies the long Euclidean-close annuli for all near-minimizers. The general weak zero-Weyl endpoint is also unchanged.

Classical inputs are DeTurck's strictly parabolic formulation, smooth parabolic existence and interior estimates, Yang's localized curvature evolution, and the Sobolev inequality on a fixed annulus. The localized continuation, collar freezing, matching construction, and their application to the actual surgery problem are proved above. This record makes no worldwide priority claim.